RSM — Just-Physics Chain (v9.3)

The explicit register. Physics is not a rendering of this framework — physics is what the explicit register studies, and this chain records where the two happen to wear the same shape. $::$ *means maps-to and never is. The implicit register fixes structure and fixes no parameters, so no quantity here is owed by the framework and none confirms it. Where an entry strains, the strain is stated with it.*

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1. Revolution, and what it organizes

The implicit register now forces revolution rather than positing it. B_n does not obtain, so the two mode-dominant regions of a frame have no path between them; the relation between them is carried by G_n revolving about B_n, which realizes the mode-swap as motion. This is the one structural fact in the chain with real reach into physics, because it makes a single question askable of any system: **what is the circulation doing?**

Three answers, and they sort a surprising amount of physics:

Circulation	Physical régime
Proceeds	duration, persistence, ordinary matter
Fails	horizon interiors, collapse
Unselected	superposition

The sorting is the entry. Nothing here says circulation *is* energy, or *is* time — those would be identity claims, and $::$ is not identity. What the table records is that three physical régimes distinguished on quite different grounds turn out to differ in one structural respect.

Where it would fail: a physical régime that is none of the three and is not a mixture of them; or a demonstration that the three are distinguished by something with no structural counterpart.

One thing the corpus material cannot bring with it. Earlier development treated rotation as a third component alongside the two modes — a Z axis, read out of 二生三. The framework does not have three modes: 三 is the pair with its bond, not a third axis. Revolution is what a frame *does*, not a part of what a frame *is*, and every entry below is written on that reading.

2. Time

The temporal case is the closest structural fit in the chain, and it comes from the implicit side rather than from analogy. Where before equals after there is no time — that is a frame's own collapse, not a value within it. So the temporal frame's balance line does not obtain, and its paradox — **the present** — is not a location the frame contains. **P_t :: now.** It cannot be occupied, only circulated; any attempt to hold it has already fixed it as past.

What follows has the right shape without needing a fourth dimension: duration is not a coordinate but what circulation around an unoccupiable centre amounts to from inside. The escapement is the clean instance — a clock's escapement does not measure time, it rocks against collapse in either direction, and the mechanism advances precisely because nothing resolves. Finer clocks do not measure time more finely; they resolve the same unoccupiable centre at a tighter radius.

The commitment this carries, stated: it types before-ness and after-ness as two conjugate magnitudes with a conserved product. That is a real claim and not a small one — it says the temporal frame is a frame in the framework's sense, not merely something the framework can describe.

Where it would fail: a physical account in which a present instant is occupiable, or in which temporal ordering survives the collapse of the before/after distinction.

3. The constants cluster

1_n :: $\hbar/2$. The complementarity floor (σ_a·σ_b ≥ $1_n/2$) and the uncertainty relation (Δx·Δp ≥ $\hbar/2$) share inequality-shape and units: the conserved product of conjugate spreads carries the dimensions of action. The shared saturation structure is not a third ground — the framework's floor is saturated by the minimal traversal, which lives in the signed chart, so on that axis the correspondence runs between a physical result and a result about the framework's chart. Mechanism note, load-bearing: quantum mechanics derives its floor from Fourier duality and noncommutativity; the framework from floor plus symmetry plus winding — no transform, no noncommutativity. Same shape, different engines, and the difference is the finding. **Where it would fail:** a physical conjugate pair with a product floor untraceable to any exclusion structure; or the winding-measure premise shown underivable.

The slice invariant :: c. The split action e^{jϕ} is the algebra of Lorentz boosts — SO(1,1), hyperbolic rotation preserving Q_j — and the boost is a rescaling of the conjugate pair at fixed invariant. **The null cone Q_j = 0 :: the light cone**; the unreachable terminals :: no massive body reaches c. A boundary, not a speed, with an algebraic backbone. **Where it**

would fail: any demonstration that the boost group's structural source requires more than the two-slice identity supplies. Named strain: the slice gives signature (1,1); physics has (1,3), which the dimension question inherits.

4. $E = mc^2$

E and m are not a conjugate pair — one conserved quantity in two unit systems is not a trade, and that reading is refused. The typing that holds: **m :: the frame unit** (1_n :: $(mc^2)^2$), **(E, pc) :: the split coordinates**, and **$E = mc^2$ :: the Amplitude Floor** — $E \geq mc^2$, equality exactly at rest. The Amplitude Floor is the boundary of the mode-expressible domain rather than a limitation on it, and $E \geq mc^2$ is a floor of exactly that kind. The typing extends: **m = 0 is $Q_j = 0$ — massless :: the null cone.** Photons live on the cone because a zero-unit frame has no paradox and no rest; the null case is the degenerate limit, not an excluded terminal. The rest-state ray $E/m = c^2$ :: the lineage line, threading the paradoxes of all particle-frames, root-capped. **Where it would fail:** a massive system saturating below its floor; or a massless system exhibiting a rest frame.

5. Wick and the two slices

$L = T - V$ :: Q_j (the difference form; the gradient slice); $H = T + V$ :: Q_i (the sum form; the circle slice); Wick rotation :: the slice exchange — $e^{iS/\hbar} \leftrightarrow e^{-\beta H}$. Physics treats the rotation as a computational device; the rendering reads it as the two-slice identity wearing spacetime signature — and the circle slice belongs to the rendering on the framework's side too, so the correspondence runs between two things of the same type. The variational principle enters modally, not teleologically: the stationary path is not chosen; it is where contributions fail to cancel. **Where it would fail:** obstructions to the continuation in curved or interacting settings that the two-slice identity cannot carry.

6. Complementarity; spin

One law, two exclusive readings, coinciding only at P_n :: Bohr complementarity — with the addition Copenhagen declined: what the two descriptions are readings *of*.

Superposition as unselected circulation. The joint read as potential is a ring of orientations, and what obtains is one selection. That is the same shape as a state whose circulation has no committed angle. Held light, and the reason is worth stating: the ring is the framework's account of *branching*, not of measurement, and nothing in it selects an observer.

The two returns :: the spinor double cover. Invariant at π, position at 2π on the framework side; sign at 2π, state at 4π for spin-½. The framework side belongs to the rendering, so this relates a physical structure to a chart feature. **The debt:** the 2π phase of spin-½ is measured. The sign is orientation-of-reading, and that measurement is owed an account. **Two candidate accounts, possibly one.** First, from the current math chain: the parturition map's two preimages are one branch compared from the child's side — an indexed comparison performed across a frame boundary — so a measured sign would be evidence of a comparison made from a register the measurement sits inside. Second, recovered from the earlier corpus, which already contained it: once the dimension route lands at three, the space of orientation-preserving frames in three dimensions is doubly connected — a closed 2π turn of frame is not contractible, while 4π is — and this is a fact about orientation itself, surviving without any metric. On that account the measured phase is the signature of orientation-space's double-connectedness, which the framework reaches through the sphere route rather than assuming. The two candidates rhyme rather than compete: the parturition realization $w = z^2$ is itself two-to-one, the model double cover. Recorded as the shapes an account would take, not as claims. **Where it would fail:** the two returns shown to be parametrization artifacts.

7. Excluded cores, and circulation failure

Circulation organized around an unoccupied core: the free vortex ($v \cdot r$ = const, with the constitutive guard that nothing was removed from the core — the exclusion is not historical), and orbital structure.

Horizon interiors :: circulation failure. Where circulation cannot proceed, a frame has no way to relate its two mode-dominant regions, and there is no in-frame path to substitute. Physical readings that follow without extra machinery: an interior is not a place the exterior fails to see into but a region no exterior frame can orient; a horizon is a threshold of that failure rather than a wall; radiation at the boundary is boundary instability rather than interior emission. The manifold-excision reading — the singularity as excised point, an unmet origin rather than an early event — is the strongest general-relativistic face of it. Anything stronger than this is the banned class.

Information as non-orientable rather than destroyed. If what is lost is the ability to orient rather than the structure itself, the disjunction between preserved and destroyed is the wrong one. Held light and flagged: this is the entry most likely to be a restatement of the puzzle in new vocabulary, since "non-orientable" here does no work the word does in geometry.

The s-orbital strain, kept visible. Quantum mechanics places the electron's density maximum *at the nucleus* for every s-state — the centre, in physics' own best description, is not spatially avoided there. Either the origin's exclusion is not positional in the atomic rendering, or this whole correspondence class is weaker than it looks. First-order strain against the framework's most important object; the exposure is the entry.

8. Frames and the lineage :: effective theories and the renormalization group

A frame :: an effective theory; 1_n :: the theory's own cutoff-relative unit; the lineage parameter :: the RG scale; index-relationality :: scheme-dependence. Physics discovered frame-by-frame description on its own: each scale its own degrees of freedom, no final theory required downward. Directionality flag: RG flow runs child→parent (coarse-graining); generation runs parent→child; the maps are inverse and must be stated as such. Named strains: RG flow is continuous, frames constitutively discrete; UV fixed points are terminals, which the framework denies — if a physical fixed point is a genuine end of the lineage, the rendering breaks there. **The Planck floor, stated as the disagreement it is:** if reality has a minimum length, P1's antecedent is false *as physics*, and every within-physics rendering loses its referent while the conditional stands. The closest thing the framework has to a falsifier-shaped exposure.

9. Irreversibility

Generation is irreversible — reversal would require P_0 to obtain — :: the thermodynamic arrow. Mechanism note as the entry: the physical arrow is statistical, the structural one modal; probability and impossibility are different modalities, and the mapping is between directions, not reasons. **Open exposure, the two arrows:** parturition-irreversibility and frame-scale shear are rival renderings, unresolved; the break-test is within-frame entropy increase with no generation, which thermodynamic relaxation appears to supply. The oldest live physics question in the project. A third rendering exists in the earlier material — entropy as loss of circulation coherence — and it is not adopted here, because it would make the second law a claim about the framework's own machinery rather than a correspondence to it.

10. The Bekenstein sort

A naive cross-frame accounting gives capacity scaling with volume; holography says area. The naivety is identifiable: a volume count comes from *packing* — children fitting inside a parent, capacities adding across what is contained. That requires children to be smaller parts of parents, which the framework denies. A branch is not a piece of a trunk, a tributary is not a piece of a river, a capillary is not a piece of an artery; each pair is one logic under different parameters, and none permits the containment count. The volume answer was never the framework's result.

Branching systems do exhibit cross-scale laws, in the form of a conserved quantity distributed across a fork rather than accumulated in a region — but the exponent varies by system, because each conserves something different. An exponent is explicit-register content belonging to an instance. The framework owes none and would gain nothing structural by having one.

The exposure that survives: **whether the framework has any quantitative cross-frame relation at all.** Its cross-frame relation is constitutive — pattern and reference, nothing passing — and conservation laws across forks arise where something passes. If nothing does, the framework makes no contact with holography in either direction: not threatened, and not testable there. A real limit on reach, recorded as such.

11. Quasiperiodicity

If the cross-frame ratio is irrational, lineage traversal winds incommensurately — return without repetition :: quasicrystalline order, orbital resonance structure. Held light, and conditional on that relation existing at all.

Predictions

1. **Unitarity as the floor's dynamical face.** The framework's floor forbids distinction-destruction; objective-collapse theories predict its violation. A confirmed objective collapse breaks the rendering — a genuine kill-condition.

2. **Decoherence at circulation failure.** If horizon interiors correspond to circulation failure, coherent processes should degrade approaching the threshold, and the degradation should not be attributable to tidal effects alone. Currently unseparated from standard predictions; the separation is owed before this counts.

3. **Structural recession** (unbuilt): the distinguishing observational signature of frame-generation against standard cosmology's dark-energy account — named as owed, not claimed.

4. **Power-law nesting** (unbuilt): scale-distribution structure in nested systems; operationalization owed.

What is not here

Deferred: gravity as recursion-geometry. Substantial earlier work exists — curvature as the cost of holding paradox open, geodesics as circulation, mass-energy and curvature as co-emergent rather than causally related — and none of it supplies what is missing, which is a route from the skeleton to *sourced* curvature. The correspondences are developed; the derivation is absent, and nothing enters without one. A warped-frame geodesic construction is the natural place a route would start.

No entry: consciousness. Nothing in the skeleton selects experience, and the earlier consciousness-energy material does not change that.

Refused typing: $E = mc^2$ as conjugate trade. **Retired:** charge conjugation, carried while the signed register was open. **Parked:** dark matter, dark energy.

Excluded class: additive conservation laws as renderings of the gradient — a sum admits terminals, and listing energy budgets under it would erase the class-selection argument.

Not carried over from the earlier physics corpus: any action functional taking a gradient of P_0, which treats an impossibility as a field; any typing of P_0 as a physical state — superposition, entropy potential, mass-energy potential — since P_0 is a condition that fails to cohere, not a state that fails to obtain; per-domain assignments of i, π and e, which are substitution rather than correspondence; and quantitative claims of the form $E \propto 1/r$ presented as derivations, which are explicit-register content and owed to an instance.

Banned: the Big Bang identification. The branching is not an event; time is frame-internal; a date belongs inside the frame it would explain. The permitted maximum: read from inside by cosmology, the frame boundary renders as a beginning.

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v9.3 — the chains state the framework as currently known; the editorial record holds how it got here.